

Graphical Abstract

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Highlights

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- Research highlight 1
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Abstract

Keywords: hydrogen, shipping, cryogenics

Nomenclature

Roman Letters

A	Sloshing excitation amplitude	m
c_p	Specific heat capacity at constant pressure	$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$
T	Temperature	K

Greek Letters

χ_{eq}	Equilibrium parahydrogen composition
χ_{ortho}	Orthohydrogen composition
χ_{para}	Parahydrogen composition

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χ_{VCS}	Dimensionless VCS position within insulation	
ω_{PO}	Dimensionless PO position along VCS	
θ	Azimuth angle	rad
θ	Zenith angle	rad

Subscripts

$evap$ Evaporation

i Inner

iw Inner wall

o Outer

ow Outer wall

VCS, i Inner side of VCS

VCS, o Outer side of VCS

1. Introduction

The growing need for decarbonisation of hard-to-abate industries has led to recent interest in liquid hydrogen (LH₂) as a high energy density, zero emission energy carrier. LH₂ has the potential to serve as a sustainable energy maritime export vector but must overcome key challenges to minimise boil off losses and tanker cool-down times in the outbound export journey to become economically viable.

The effective delivered price of LH₂ is dependent on the total boil-off losses incurred along the supply chain. For a fixed LH₂ production cost, the magnitude of total boil-off losses proportionally increases the effective delivered price of LH₂, as additional LH₂ would have to be purchased to counteract the boil-off losses. Over NASA's space shuttle program, 45% of LH₂ purchased was lost to boil-off [1] and meant that the effective delivered LH₂ price to the shuttle was approximately double the purchase price. 12.6% of the total purchased LH₂ was lost during static storage tank filling and

12.2% was lost due to evaporation within the tank [1]. Minimising the boil off from these sources would be critical to support the scale up of LH₂ tanks for energy export, where low cost H₂ is critical for the business case.

the time taken to fill and cool-down LH₂ tankers is another critical factor on LH₂ export viability. Apart from the voyage of the Suiso Frontier pilot carrier, no large scale commercial shipping of LH₂ has yet occurred. However, insights into potential operational challenges can be made by considering the bulk shipping of liquefied natural gas (LNG). While operating at a higher temperature ($\sim 111\text{ K}$), the low storage temperature of LNG presents certain challenges which may also be present for LH₂. Under IGC regulations, it is necessary to inspect ship-borne LNG tank every 5 years to check for damage. This requires warming the tanks to ambient temperature to allow human inspectors to enter, and upon inspection, a pre-cooling step must be conducted before filling can commence. Since the time spent in dry-dock should be minimised, it is important to minimise the duration of this pre-cooldown stage as well as any boil-off losses incurred. For large LNG membrane tanks, this process typically takes 8 – 12 hours [1-3].

Compared to LNG tanks, large scale LH₂ tanks are likely to have greater insulation thicknesses as well as different insulation materials. Furthermore, at temperatures close to $\sim 20\text{ K}$, material thermophysical properties such as thermal conductivity and specific heat capacity are likely to be significantly different compared to at $\sim 111\text{ K}$. Previous experience with vacuum-insulated LH₂ tanks initially at ambient temperature suggest long time scales for pre-cooling, particularly with vacuum insulated tanks. Liebenberg (1965) [2] outlined the approaches taken for the chill down of a 208 m^3 doubled-walled spherical LH₂ tank, initially at ambient temperature. Loading of the tank was conducted over the span of one week. In one case, two batches of 30 m^3 of LH₂ were loaded two weeks prior to pre-chill the tank. This reduced the overall boil-off rate from 15.5 to 7.5 %. After 3 weeks, the daily boil-off rate was observed to be 30 % higher than normal, though this decreased to 22 % after 5 weeks. A similar observation was made during the chill down of a $3,400\text{ m}^3$ spherical LH₂ tank which was being returned to service and was initially at ambient temperature [3]. Approximately $1,500\text{ m}^3$ of LH₂ was loaded over a period of 9 days. An average boil-off rate of 0.045 % per day was observed over the following 2 months, significantly higher than the normal rate of boil-off.

One path to minimise the boil off during LH₂ tanker export is the use of improved thermal insulation materials such as glass bubbles. For example,

NASA’s use of glass bubbles in a 5700 m^3 tank is targeted to achieve a nominal boil off of 0.048 %/day compared to the measured 0.075 %/day boil off achieved in a perlite insulated 3407 m^3 tank [1]. However, this approach requires high vacuum levels and may not be practical for use in export scale spherical LH₂ tanks reaching 40000 m^3 capacity. Additionally, the thermal mass of improved insulations may still have appreciable filling boil off losses and cool down times. An alternative method to reduce the boil off and cool down times of maritime LH₂ export is to utilise a Vapour Cooled Shield (VCS) in the annulus insulation layer. In this concept, a stream of boil off gas is passed through the insulation to intercept heat leak while also providing a second direction of cooling for the purposes of tank cool down. Figure 1 shows the VCS premise which typically utilises a coil or series of tubes that pass re-gasified LH₂ through the insulation layer to lower the tank’s radial temperature gradient and thus net boil off rate.

Employing a VCS could reduce the net boil off rate during all stages of the LH₂ maritime export journey, however the main focus of this initial study was the outbound journey. The primary sources of boil off result from filling of the empty, warm LH₂ tank, the transient cool-down of the tank insulation once full of LH₂ and the steady state boil off once the tank insulation reaches thermal equilibrium as shown in Figure 2. Evaluating the performance of a VCS in these applications requires assumptions of the initial tank conditions for LH₂ filling which are taken as empty and a uniform temperature of 300 K across the entire tank for simplicity. The transient boil off process commences once the inner wall has reached 21 K for the accumulation of saturated LH₂ while the outer insulation is at a higher temperature. While the steady state boil off stage occurs once the tank insulation reaches thermal equilibrium and the boil off reaches a pseudo-steady state. The primary focus of VCS development has been for steady state boil off reduction though the investigation of VCS performance for the filling and transient stages could indicate broader VCS utility.

VCS LH₂ tanks only employing the sensible cooling of H₂ have been experimentally demonstrated to reduce the static boil-off rate by 35% [4] while modelling of new VCS designs indicates that the boil-off rate could be reduced by approximately 60% [5, 6, 7]. Additional boil off reduction is also possible by inducing the endothermic parahydrogen to orthohydrogen (P-O) conversion through installing a catalyst powder bed within the VCS tubes. The specific energy adsorbed by both sensible and catalytic cooling are shown in Figure 10 with both being significant compared to the latent heat of

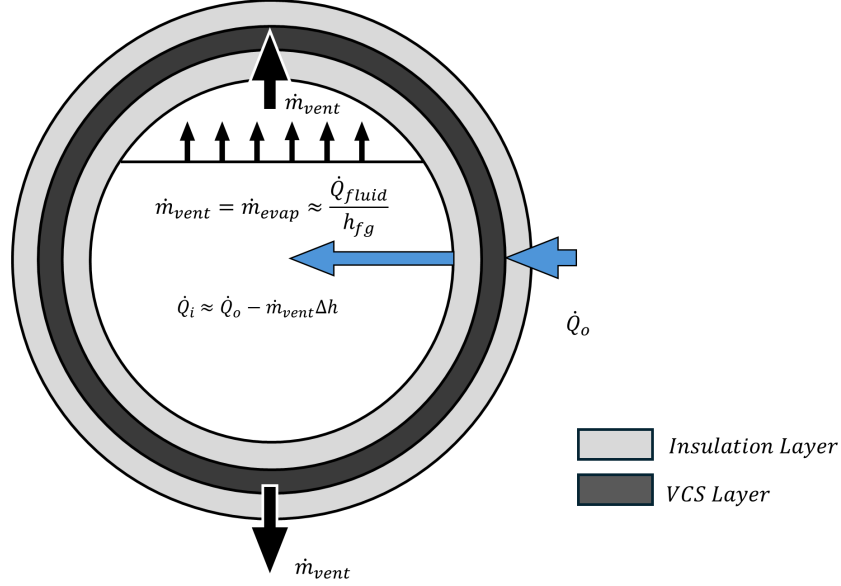


Figure 1: Illustration of the cross section of a LH_2 tank employing a VCS. Under theoretical steady state conditions and a sufficiently high fill level, the rate of venting ($\dot{m}_{vent}$) can be assumed to equal the rate of evaporation ($\dot{m}_{evap}$), which can be assumed to be proportional to the steady state rate of heat transfer ($\dot{Q}_i$) into the tank. Heat carried away by the VCS ($\dot{Q}_{VCS} = \dot{m}_{vent} \Delta h$) can be assumed to be equal to the difference between $\dot{Q}_i$ and the rate of external heat transfer ($\dot{Q}_o$).

evaporation of liquid parahydrogen at 446 kJ/kg . Studies indicate that $\text{P} \rightarrow \text{O}$ conversion could further reduce the steady state heat flux compared to a non-catalytic VCS by approximately 10% [5, 7, 8]. The P-O contribution to VCS heat flux reduction is significantly lower than theoretically indicated as the VCS performance is dependent on the radial temperature gradient created by the VCS. Overall, the required volume of catalyst and catalytic VCS design considerations are not widely reported, posing challenges to progress the development of catalytic VCS tanks for minimum LH_2 waste.

Evaluating the performance of a VCS for a 40000 m^3 export scale tank in the above scenarios requires a mathematical model to couple heat transfer resistance network and kinetic model of P-O conversion together to quantify the radial temperature gradient and resultant net LH_2 boil off rate. This paper investigates several modelling assumptions on VCS performance while

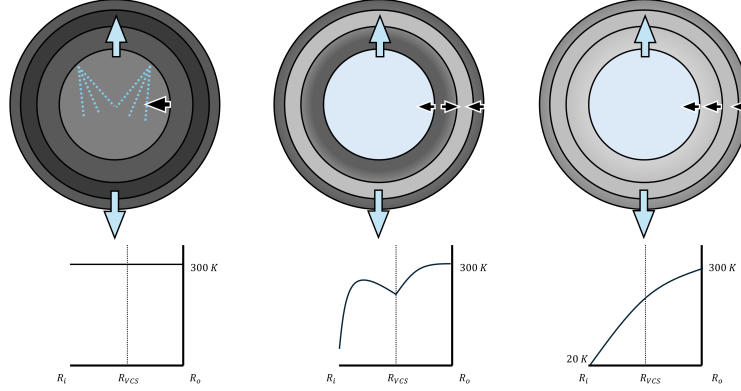


Figure 2: Illustration of a VCS LH₂ tank in the sequential stages of the LH₂ export process. Stages depict LH₂ spray chill-down of the ambient temperature tank, the transient cool down of the tank once full of LH₂ and the steady state boil off at thermal equilibrium respectively. The corresponding insulation radial temperature profiles are also shown.

also quantifying the utility of a VCS to reduce the boil off rate during static and dynamic tank filling scenarios.

1.1. Theoretical Background

Steady State

The utility of a VCS and VCSPO for boil-off reduction can be understood by first considering a non-VCS storage tank. For a spherical cryogenic storage tank under quasi-steady state conditions, the rate of heat ingress and boil-off can be expressed by Eq. 1.

$$\dot{Q}_{ss} = 4\pi k_{eff} \frac{T_{w,i} - T_{w,o}}{1/R_i - 1/R_o} \quad (1)$$

Where $T_{w,i}$ and $T_{w,o}$ refer to inner and outer wall temperatures respectively, R_i and R_o refer to inner and outer wall radii respectively, and k_{eff} refers to the effective thermal conductivity of the insulation layer for the given Cold Boundary Temperature (CBT) and Warm Boundary Temperature (WBT) under steady state conditions. For simplicity, it can be assumed that $T_{w,i}$ and $T_{w,o}$ are 20 and 300 K respectively. If negligible thermal strati-

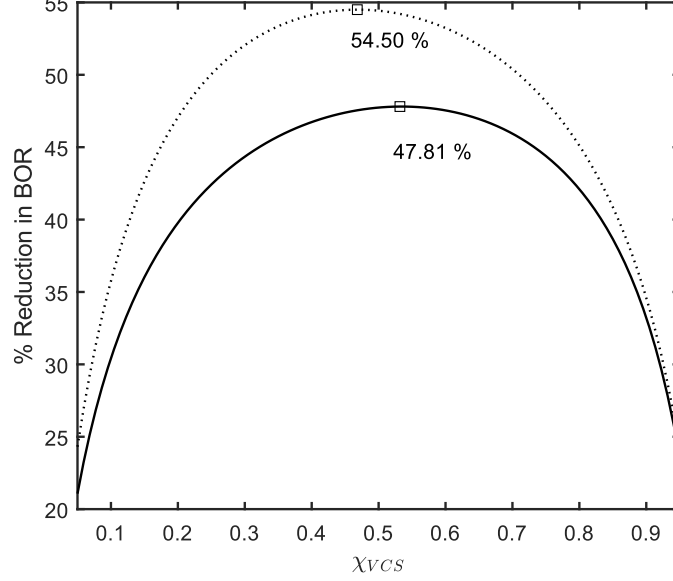


Figure 3: Theoretical reduction in steady state BOR for the 40,000 m^3 spherical VCS (solid line) and VCSPO (dotted line) considered in this study, as a function of χ_{VCS} , when thermal conductivity is assumed to be independent of temperature. Squares correspond to the 'optimal' position of the VCS where reduction in BOR is maximised.

fication in the liquid and vapour is assumed, the boil-off rate can be expressed as Eq 2.

$$BOR_{ss} = \frac{\dot{Q}_{ss}}{m_l h_{fg}} \quad (2)$$

Where h_{fg} refers to the enthalpy of vapourisation and m_l refers to liquid mass.

If a VCS is introduced at position R_{VCS} within the insulation layer (as shown in Figure), the reduction in the rate of heat transfer can be estimated by making several simplifying assumptions. If the VCS is assumed to be of a uniform temperature, the rate of heat transfer from the VCS to the inner wall ($\dot{Q}_{VCS,i}$) can be expressed as:

$$\dot{Q}_{VCS,i} = 4\pi k_{eff,i} \frac{T_{VCS} - T_{w,i}}{1/R_i - 1/R_{VCS}} \quad (3)$$

Similarly, the rate of heat transfer from the outer wall to the VCS ($\dot{Q}_{VCS,o}$)

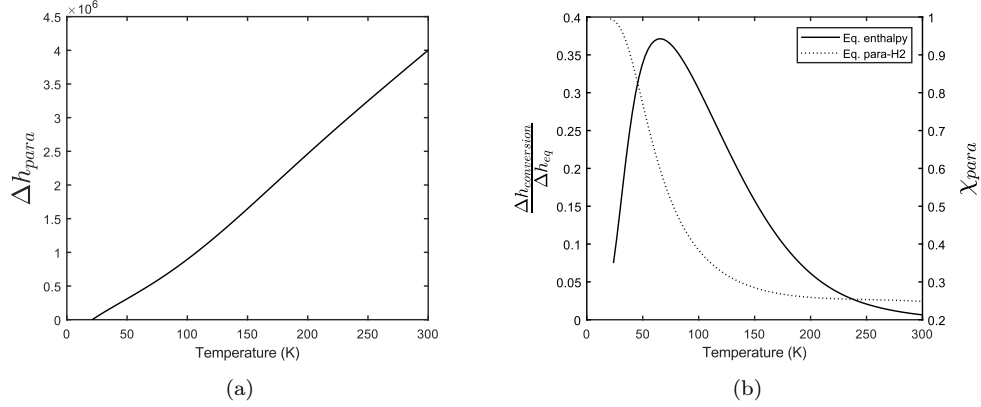


Figure 4: Cooling potential of gaseous hydrogen from 20 K to 300 K. a) Change in 'sensible' enthalpy of para-H₂; b) contribution of para-ortho conversion to change in total enthalpy of hydrogen, assuming para fraction is equal to equilibrium fraction at the corresponding temperature.

can be expressed as:

$$\dot{Q}_{VCS,ss} = 4\pi k_{eff,o} \frac{T_{w,o} - T_{VCS}}{1/R_{VCS} - 1/R_o} \quad (4)$$

Performing an energy balance on the VCS, under steady state conditions, a relationship between $\dot{Q}_{VCS,o}$ and $\dot{Q}_{VCS,i}$ can be defined.

$$\dot{Q}_{VCS,o} = \dot{Q}_{VCS,i} + \dot{m}_{BOR}(h_{exit} - h_{vent}) \quad (5)$$

The rate of venting can be expressed as $\dot{m}_{BOR} = \dot{Q}_{VCS,i}/h_{fg}$. For simplicity, it can be assumed the hydrogen entering the VCS is equal to the inner wall temperature ($T_{vent} = T_{w,i}$) and the hydrogen exiting the VCS is to equal the VCS temperature ($T_{exit} = T_{VCS}$). Similarly, since CBT and WBT are fixed, the effective thermal conductivity of the insulation on the inner and outer layers can be expressed as a function of VCS temperature (T_{VCS}). Hence, both sides of Eq 5 can be expressed as a function of T_{VCS} .

If k_{eff} is assumed to be independent of temperature (ie: fixed thermal conductivity), and the specific heat capacity of the venting and exit gases is assumed to constant, then the optimal location of the VCS (χ_{VCS}) can be shown to be vary between 0.52 to 0.54, leading to a reduction in steady state BOR of $\sim 48\%$ per day (see Figure).

If a para-ortho converter is installed in the VCS, the effect can be quantified by modifying h_{exit} in Eq. 5 to reflect the additional enthalpy change

due to spin isomer change. If it is assumed that the ortho- fraction of the hydrogen leaving the VCS is equal to the equilibrium ortho- fraction at the VCS temperature, then the reduction in steady state BOR increases to 54.5 % (see Figure) at an optimal χ_{VCS} of 0.45 to 0.47.

In reality, thermal conductivity varies strongly with temperature. Furthermore, temperature profile of the VCS is unlikely to be uniform and the conversion of parahydrogen along the length of the VCS will depend on a range of factors such as catalyst activity, flow rate and piping geometry.

Tank cooldown

In theory, the minimum quantity of LH₂ required to cool a given mass from ambient temperature to the saturation temperature ($\sim 20\text{ K}$) can be expressed as:

$$m_{sacrificial} = m_{tank} \int_{T_{sat}}^{300} \frac{c_{tank}(T)}{h_{eq}(T) - h_{inj}} dT$$

Where $c_{tank}(T)$ refers to the specific heat capacity of the material at temperature T , h_{inj} refers to the enthalpy of the injected LH₂ and $h_{eq}(T)$ refers to the enthalpy of hydrogen at temperature T , at its equilibrium spin isomer fraction.

For the inner wall of a 40,000 m^3 spherical tank with a thickness of 40 mm made from 304 stainless steel, this is calculated to be 91.5 tonnes-LH₂, assuming LH₂ is initially injected as a saturated liquid at atmospheric pressure and leaves the tank at atmospheric pressure equal to tank temperature.

In reality, gaseous hydrogen is likely to be vented initially at a temperature significantly lower than the average wall temperature.

1.2. Literature Review

Multiple numerical modelling studies have explored the utility of a VCS and VCSPO for the reduction of steady state BOR in LH₂ tanks. Primary variables include the relative position of the VCS in the insulation layer from the inner wall, the LH₂ tank modelling assumptions and the dimensional complexity of the model. This understanding framed the development of a high-fidelity combined heat transfer and kinetic model to accurately quantify the performance of a VCSPO in reducing the net boil-off rate and chill down time in a LH₂ tank.

Modelling by Shi et. al. [5] found that a VCS positioned 50 % through the annulus insulation from the inner wall ($\chi_{VCS} = 0.5$) reduced the local temperature from $T = 215$ to 128 K leading to a 61 % reduction in the net heat leak to the LH₂. Xu et. al. [7] observed a similar model result where a VCS position at $\chi_{VCS} = 0.4$ reduced the local temperature from $T = 186$

to 106 K leading to a 56 % reduction in the net heat leak to the LH_2 [10]. $P \rightarrow O$ conversion acts to further decrease the VCS temperature leading to reduced heat leak into the LH_2 . Meng et. al. [9] found that $P \rightarrow O$ conversion could reduce the VCS temperature from $T = 92$ to 85 K leading to a 10 % net heat leak reduction compared to a non-catalytic VCS. While the $P \rightarrow O$ conversion can represents 20 % of the cooling done by the VCS [5], its effect on the net boil-off rate is diminished as the conversion only results in a small temperature gradient change.

Optimising the boil-off reduction possible by the VCS depends on its relative position through the insulation layer to balance the heat leak from the outer wall to the VCS and from the VCS to the tank inner wall. Moving the VCS closer to the inner wall increases the heat leak into the LH_2 due to a larger temperature gradient from the VCS layer. Whereas placement of the VCS closer to the outer wall increases the net heat leak from the outer wall to the VCS due to a larger temperature gradient. This impacts the temperature of the VCS and thus the temperature gradient from the VCS to the inner wall. This behaviour means an optimal VCS radius exists to minimise the net boil-off rate and can be found through modelling studies. Shi et al. [5] found placing a non-catalytic VCS at $\chi_{VCS} = 0.5$ minimised the LH_2 boil-off rate while a position of $\chi_{VCS} = 0.43$ minimised the net boil-off rate in a catalytic VCS for a multi layered insulated tank. This optimal VCS position was consistent with other VCS studies on multi layered insulation LH_2 tanks [7, 10]. However, the optimal VCS position is influenced by the selected insulation as Zheng et al. (2023) [6] found a $\chi_{VCS} = 0.3$ optimal position for a LH_2 tank using hollow glass microsphere insulation due to its different thermal properties.

The proposed 40,000 m^3 LH_2 tank of this study is anticipated to utilise bulk packing insulation materials such as perlite or glass microspheres under vacuum [11]. However, the thermal conductivity of glass microspheres has had limited characterisation with respect to temperature other than testing at $T = 77$ and 90 K [12, 13] which poses challenges for accurately assessing VCS performance. Accurately modelling the VCS performance requires high resolution characterisation of the insulation thermal conductivity with respect to temperature as the VCS position changes the boundary temperatures and means the use of an effective thermal conductivity is not appropriate. For these reasons, helium filled polyurethane foam (PUF) was selected as the insulation material for the VCS model in this study. The thermal conductivity of PUF is well characterised with respect to temperature [14] and

is a material commonly used in cryogenic insulation for liquefied natural gas tanks of equivalent size to this model [15]. The baseline LH₂ boil-off rate is anticipated to be higher than an equivalent tank using glass microspheres as the thermal conductivity of PUF is $33.6\times$ larger than the glass microspheres at $T = 77$ K under high vacuum [12]. Modelling studies indicated that a $40,000\text{ m}^3$ LH₂ tank with PUF insulation to have a 0.345 %/day boil-off [15] while a 20000 m^3 LH₂ tank with glass microspheres is anticipated to have a 0.06 %/day boil-off rate [16]. The higher baseline boil-off in a PUF tank represents a scenario where a VCS could offer significant boil-off reduction to potentially achieve a similar net boil-off as a glass microsphere insulated tank without the requirements for a vacuum layer.

The performance of a VCS system is assessed through heat transfer resistance network models established in 1 dimensional (1D) and 2 dimensional (2D) networks. These models have different assumptions on the temperature of the VCS which in turn impact the estimated net boil-off reduction as shown in Table 1. 1D models characterise the heat flux in the radial direction of the tank but assumes that the VCS is a uniform temperature in the polar direction for simplicity as shown in Figure 3. 2D models characterise the heat flux in both the radial and polar direction to capture how the H₂ heats up along the VCS flow path from approximately $T = 20$ to 120 K [11]. The VCS temperature directly impacts the heat flux through the temperature gradient and indirectly through the local insulation thermal conductivity. The 1D model also assumes that $\text{P}\rightarrow\text{O}$ conversion reaches its equilibrium p-H₂ fraction at the VCS temperature whereas the 2D model allows the use of a catalyst volume and kinetic model to evaluate the extent of conversion. The different assumptions on the temperature profile also impacts the maximum p-H₂ fraction change as the temperature of the catalyst bed in the 2D model dictates the maximum possible composition change rather than the outlet temperature of the VCS as in the 1D model.

The impact of the VCS model dimensions on the $\text{P}\rightarrow\text{O}$ contribution to reducing the net heat flux are highlighted in Table 2. 1D models generally achieved a 10 % $\text{P}\rightarrow\text{O}$ cooling enhancement with a 100 % conversion extent. An optimal VCS position of approximately $\chi_{VCS} = 0.5$ has been shown to minimise the net heat leak while maximising the $\text{P}\rightarrow\text{O}$ contribution to the heat leak reduction suggests an optimal $\chi_{VCS} = 0.4$ [7] though this is dependent on the insulation selected. Most 1D models assume 100 % extent of conversion while the Lv et al. model [10] used a correlation to estimate the $\text{P}\rightarrow\text{O}$ extent of conversion as 78 % which resulted in the relatively smaller

Table 1: Overview of large, land-based LH₂ dewars. As the time of writing, the 5,250 m³ LH₂ tank at KSC has not been fully commissioned, and hence, BOR figures are only theoretical and may not reflect the observed BOR.

Author	Model Type	Optimal VCS location (χ_{VCS})	Additional BOR reduction due to P→O converter (%)	VCS Temperature (K)	P→O conversion (%)
Shi et al. (2023) [5]	1D	-	11	120	100
Lv et al. (2024) [10]	1D	0.5	5	140	78
Meng et al. (2023) [9]	1D	0.4	10	106	100
Xu et al. (2023) [7]	1D	0.5	12	106	100
Leng et al. (2024) [8]	1D	0.4	7.4	100	100
Leng et al. (2024) [8]	2D	0.4	13.3	20 - 120 K	100

P→O cooling enhancement. Whereas the Leng et al. 2D model [8] accounted for the H₂ temperature increase along the VCS path, resulting in a higher P→O contribution as the outlet temperature of the catalytic converter was typically higher than that estimated by the uniform VCS temperature assumption of the 1D model. Xu et al. [7] found that modelling the P→O conversion as an isothermal process resulted in a 3.5 % decrease in the net heat flux compared to assuming an adiabatic conversion. This highlights the need to model the P→O conversion coupled with heat transfer to accurately capture the extent of isothermal or adiabatic performance of the catalytic bed. The Leng et al. model [8] showed that a 1D model underpredicted the net LH₂ boil-off rate by 14 % compared to the 2D model which matched experimental VCS results to within 1 % of experimental values. This underprediction was caused by the uniform VCS temperature assumption which impacts the heat transfer rate directly and indirectly through the local insulation thermal conductivity. These studies demonstrate that P→O cooling

contribution is sensitive to the extent of conversion and temperature of the converter. Both of these parameters cannot be accurately considered in a 1D model and thus a 2D VCS model that incorporates a kinetic model for $P \rightarrow O$ conversion is required to more accurately evaluate the $P \rightarrow O$ cooling contribution. This study utilises a 2D VCS model with a kinetic model of a $P \rightarrow O$ converter to model the $P \rightarrow O$ cooling contribution to a VCS on a $40,000 \text{ m}^3$ tank to support future VCS implementation.

In summary, the majority of studies focusing on VCS-PO There is relatively little literature on the applicability of VCS to large-scale bulk-insulated storage tanks. As far as the authors are aware, no prior studies have attempted to perform modelling on the effect of a VCS on tank cooldown time and efficiency of storage tanks.

2. Objectives

Literature indicates that the application of a VCS to a LH_2 tank has the potential to significantly reduce the static boil off rate to improve the economic viability of LH_2 export. VCS performance is influenced by the assumption of either 1D or 2D dimensions as this impacts the temperature profile and extent of P-O conversion. However, the consequences of these assumptions are not well understood for the scale and insulation type of export scale LH_2 tanks. The radial position has also been modelled to impact the net steady state boil off rate but it is unclear how the VCS position effects the boil off rate during warm tank filling and the transient cooldown of the full tank. Further modelling is required to quantify the performance of a VCS in reducing the boil off from each stage of tank use to support tank designers minimise total boil off for economically viable LH_2 export. This was conducted through this work on the basis of applying a simple VCS design to a spherical 40000 m^3 export scale LH_2 tank using PUF insulation. The specific objectives of this work were to:

- Compare the steady state boil off rate across 1D and 2D VCS models to evaluate the assumptions of VCS temperature profile and extent of P-O conversion on performance.
- Quantify the impact of the VCS radial position on steady state boil off rate.

- Investigate VCS performance for reducing the LH_2 boil off mass and duration of filling a warm, empty tank and the boil off mass during the subsequent transient filled tank cooldown process.

3. Methodology

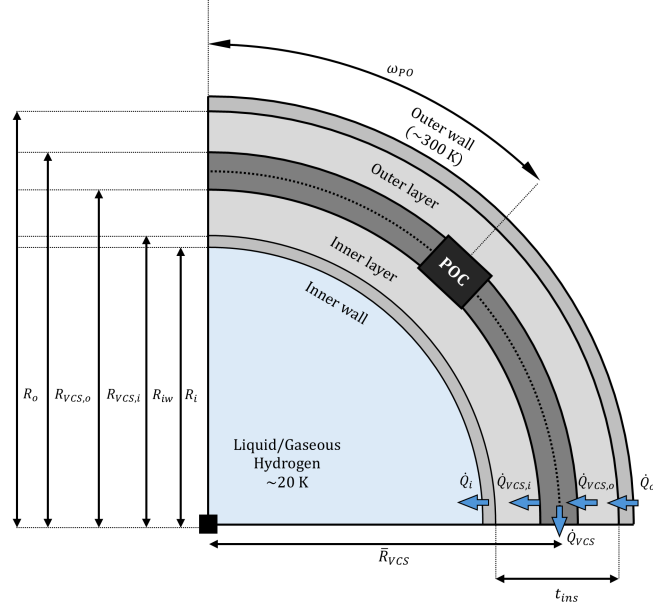


Figure 5: Illustrations of the dimensions of the VCSPO considered in this study. For simplicity, the outer wall is treated as uniform and equal to 300 K , while the inner wall temperature is treated as uniform and equal to 21 K under steady state conditions, and varying over time under dynamic conditions.

This study utilises a 1D and 2D model to estimate heat flows and temperature profiles for a non-VCS, VCS and VCSPO LH₂ tank. Since this study aims to assess the additional benefits of using a 2D model (especially in the context of P-O conversion), the 1D model primarily serves as a benchmark. 1D and 2D VCS models were compared in the application of steady state heat leak reduction while the 2D model was subsequently used to investigate the VCS performance during tank filling and the transient tank cool down.

3.1. 1D Model

The 1D model inherently makes the following assumptions:

1. There are a sufficient number of vapour-carrying tubes such that VCS and VCSPO temperature can be treated as uniform.
2. Temperature profiles in the insulation layer are spherically symmetric.
3. Negligible thermal stratification in the liquid or vapour, and inner wall temperature can be treated as uniform.
4. Outer wall temperature can be treated as uniform and constant over time.
5. For VCSPO, there is 100 % P-O conversion along the VCS tubing.

The equations for solving the 1D model are laid out in Eqs. (??). The temperature dependent thermal conductivity of PUF insulation is provided by NIST [17].

3.2. 2D Model

3.2.1. Tank and VCS

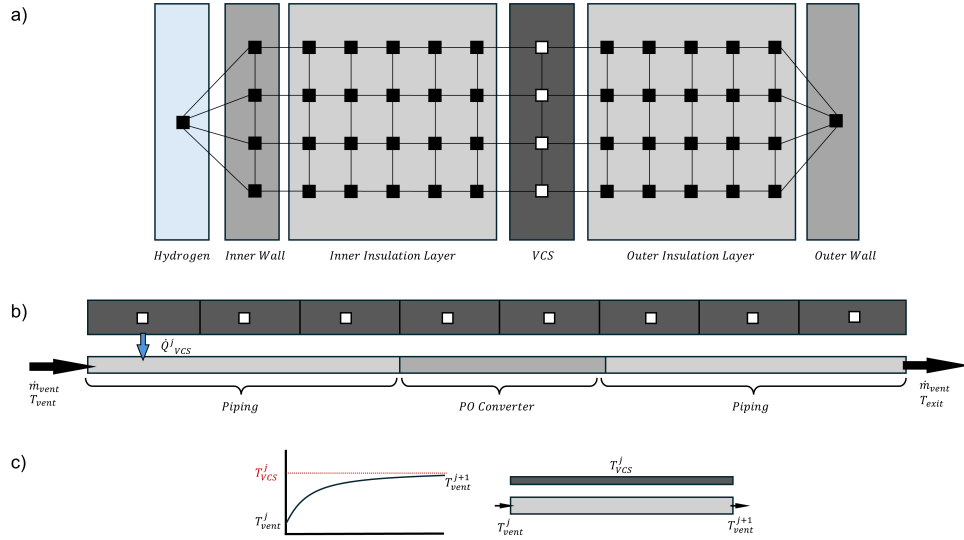


Figure 6: Overview of the 2D VCS thermal network model used in this study: a) Overview of wall, insulation and VCS sections. It is important to note that the outer wall temperature was assumed to be fixed at 298 K in all models; b) Flow of the vented hydrogen vapour is represented by a 1D pipe model, with the pipe thermal mass assumed to be negligible; c) Pipe-vapour heat transfer is treated by assuming a uniform boundary temperature along each section of the pipe.

The 2D model is used to assess both steady state and dynamic performance. Figure 6 shows the structure of the thermal network model used to

represent the tank. For simplicity, the vapour-cooled shield is approximated as a 1D network. Since the shield's thermal conductivity is significantly higher than the surrounding PUF, it was assumed temperature gradients in the radial direction would be negligible. The flow of hydrogen was assumed to be directed by a series of pipes extending from the top of the tank down to the bottom. Pipe walls were assumed to equal the temperature of the adjacent shield. In the case of VCS-PO, reactor temperature was approximated. Since the thermal gradients along the length of the PO converter

The assumptions underlying the 2D model are given as:

1. There are a sufficient number of vapour-carrying tubes such that VCS and VCSPO temperature can be treated as axisymmetric.
2. Temperature profiles in the insulation layer are axisymmetric.
3. Inner wall temperature can be treated as axisymmetric.
4. Outer wall temperature can be treated as uniform and constant over time.
5. A catalyst bed is located at the midpoint of the VCS flow path.

Importantly, compared to the 1D model, assumptions regarding uniformity of temperature in the insulation layer are relaxed. This allows for more accurate prediction of vapour heat transfer along the length of the pipe as well as PO converter performance. Energy balance in the j^{th} section of the VCS is given by:

$$\frac{dT_{VCS}^j}{dt} = \frac{1}{m^j c_v} (\dot{Q}_{VCS,o}^j - \dot{Q}_{VCS,i}^j - \dot{Q}_{VCS}^j + \dot{Q}_{cond}^j) \quad (6)$$

Where $\dot{Q}_{pipe}^j$ refers to the heat transfer from the VCS wall section to the pipe section. If the thermal mass of the pipe is assumed to be negligible, the rate of heat transfer from the pipe to the VCS can be assumed to equal the rate of heat transfer from the vented vapour stream to the pipe:

$$\dot{Q}_{VCS}^j = \dot{Q}_{vent}^j \quad (7)$$

For simplicity, heat transfer between the vapour and pipe walls was approximated by considering laminar, transitional and fully developed turbulent flow in a pipe of uniform temperature [18].

$$Nu_D = \begin{cases} Re < 2000 : & 3.66 \\ 2000 < Re < 6000 : & 0.116 * (Re^{0.667} - 125) * Pr^{0.4} \\ Re > 6000 : & 0.023 * Re^{0.8} * Pr^{0.4} \end{cases} \quad (8)$$

Reynolds number in the pipe is given by:

$$Re = \frac{VD}{\nu} \quad (9)$$

Energy balance within the inner wall is given by:

$$\frac{dT_{wall}^j}{dt} = \frac{1}{m_{wall}^j c_v} (\dot{Q}_{ins-wall}^j - \dot{Q}_i^j + \dot{Q}_{cond}^j) \quad (10)$$

where $\dot{Q}_{ins-wall}^j$ refers to the rate of heat transfer from the insulation to the inner wall.

3.2.2. P-O Converter

The 2D VCSPO utilised the first order kinetic model to describe the P-O conversion rate as a function of temperature and H₂ flow rate in a packed tube of catalyst powder. This model was selected for its reasonable accuracy over a wide temperature range [19]. The integrated form of the first order kinetic model is shown in Eq. (11). where the outlet p-H₂ fraction from a catalyst bed is dependant on the initial (X_1) and equilibrium (X_e) p-H₂ fractions, the rate constant of the catalyst (k) and the residence time of H₂ in the catalyst bed (t).

$$X_2 = X_e + (X_1 - X_e)e^{-kt} \quad (11)$$

The residence time is expressed by Eq. (12). as a function of catalyst bed volume, V , (inclusive of the voids between catalyst particles) and the volumetric H₂ flow rate through the bed at actual temperature and pressure, $\dot{v}$.

$$t = \frac{V}{\dot{v}} \quad (12)$$

The general rate constant expression of Eq. (13). was used to represent catalyst activity using the pre-exponential factor, A , activation energy, E , Ideal gas constant, R , and VCSPO temperature, T . This expression was

used based on the authors success in using it fit ferric oxide P-O conversion data over a relevant temperature range [20]. For the purposes of representing a basic VCSPO system, Ferric oxide powder was selected as the catalyst in the model as it represents a well characterised catalyst [20] with reasonable activity and is widely distributed under the trade name "Ionex type OP" by Molecular products [21]. Values of the kinetic parameters were selected to represent a conservative ferric oxide catalyst activity using $A = 1000$ bara/min and $E = 350$ J/mol and were obtained by fitting experimental conversion data [22, 20]. This represented an approximately 50% lower activity compared to the ferric oxide catalyst tested by Hutchinson when evaluated at 77 K, 2 bara [20], thus representing a conservative case. The catalyst volume in each VCSPO tube was sized to target a 95% extent of conversion based on the flow rate in the non-VCS scenario.

$$k = \frac{A}{P} e^{\frac{-E}{RT}} \quad (13)$$

The catalytic VCSPO scenario employed a ferric oxide catalyst bed at the equator of the VCS to represent a VCS 'designed for manufacturing'. The catalyst bed was placed at the equator to allow preheating of the H₂ in the VCS so that the inlet to the catalytic bed had a sizeable p-H₂ equilibrium driving force to induce conversion. The catalytic bed was discretised into 20 axial sections, where a maximum temperature gradient of $T = 1$ K was typically observed across the cells. A high resolution was required for the catalyst bed as the temperature impacted the equilibrium driving force and the kinetic parameters of conversion.

Assuming an average temperature, required approximately 650 L of catalyst for the overall boil off rate this total volume was divided into 50 VCS tubes

3.2.3. Vapour

During transient modelling of the cooldown process, it is assumed that the tank (following a purging process) initially contains parahydrogen vapour at ambient temperature. Upon injection of liquid spray into the tank, mass and energy balance within the tank

$$\frac{dm_v}{dt} = \dot{m}_{inj} - \dot{m}_{vent} \quad (14)$$

Table 2: Specifications of VCS system for the 2D model

Number of VCS tubes	50
VCS wall thickness	0.015 <i>m</i>
VCS wall material	304 stainless steel
Inner catalyst tube diameter	0.05 <i>m</i>
Relative catalyst bed position	0.5
Catalyst bed length	6.61 <i>m</i>
Catalyst bulk volume in each VCS tube	12.98 <i>L</i>
Total catalyst volume in overall VCSPO	648.9 <i>L</i>

Table 3: Design parameters of LH₂ tank used in VCS model

Inner tank volume	40000 <i>m</i> ³
Inner tank diameter	21.216 <i>m</i>
Outer tank diameter	22.070 <i>m</i>
Outer wall thickness	0.014 <i>m</i>
Wall material	304 stainless steel
Annulus thickness	0.78 <i>m</i>
Insulation material	PUF
VCS wall thickness	0.015 <i>m</i>
VCS flow diameter	0.03 <i>m</i>
VCS wall material	304 stainless steel

$$\frac{dU_v}{dt} = \dot{m}_{inj}h_{inj} - \dot{m}_{vent}h_v \quad (15)$$

where h_{inj} and h_{vent} refer to the enthalpy of the spray injected into the tank during cooldown and bulk vapour enthalpy respectively. Thermal stratification in the vapour during cooldown is therefore assumed to be negligible, with exit vapour temperature equal to average vapour temperature. In reality, this is only likely to be true if fluid flow induced by the spray is sufficient to cause mixing of the vapour. For simplicity, it was assumed constant tank pressure was maintained by a continuous rate of venting. This rate of venting was determined by an iterative solver at each time step.

The rate of heat transfer between the vapour and tank wall is assumed to occur via natural convection [23]:

$$Nu = 0.197Ra^{0.321} \quad (16)$$

where Ra refers to Rayleigh number, given by:

$$Ra = g \frac{\beta(T_{wall} - T_v)L^3}{\nu^2} \quad (17)$$

Here, β refers to coefficient of thermal expansion of the vapour and ν refers to kinematic viscosity. All thermophysical properties are taken at the average temperature of the wall and bulk fluid. Fluid properties were taken from NIST's REFPROP (version 9.1), based on Leachmann's equations of state [24].

3.3. Solution methods

A grid independence study was used to determine appropriate spatial discretisation of the tank [25], with M set to 20 and N set to 30. Within the inner and outer insulation layers, the model was discretised to provide greater "resolution" at the boundaries (inner wall, VCS and outer wall) where a higher thermal gradient was expected.

For the transient model, a simple forward time stepping method was employed with a time step of $dt = 1$ s. For the steady state model, the solution was determined by running the transient model with fixed wall temperature boundary conditions (20.26 K inner wall temperature and 298 K outer wall temperature) until differences in $\dot{Q}_i$ and $\dot{Q}_o$ fell below 0.5 %. Steady state.

4. Results

4.1. Steady state performance

The net heat leak reduction to a LH₂ tank with a VCS modelled in 1D and 2D geometry are shown in 7. The heat leak is proportional to the net LH₂ boil-off rate from the tank based on the steady state assumption with saturated LH₂ storage. Maximum heat leak reductions of 54.5 % and 58.5 % were predicted for the 1D and 2D non catalysed VCS models respectively. These results align with results from the literature which typically cite a 60 % heat leak reduction for VCS applied to MLI insulated tanks [5, 7].

Both models indicate the existence of an optimal VCS position to minimise the net heat leak into the tank. This peak primarily resulted from a decreasing spherical heat transfer resistance at closer VCS positions and an increasing temperature gradient at further out VCS positions. This can be intuitively understood based on the spherical conduction heat transfer equation (ref). The 1D and 2D models predicted similar optimal non-catalytic VCS positions of $\chi_{VCS} = 0.43$ and 0.47 respectively. This result was similar to the optimal VCS position indicated by modelling studies on MLI insulated tanks where an optimal $\chi_{VCS} = 0.5$ was found [11, 104]. However, the heat leak reduction of these studies ranged from 56 to 70 % [7, 10] indicating that the net boil-off reduction is sensitive to tank geometry and insulation, though a broad optimal heat flux reduction peak exists.

Similar relationships of heat leak reduction with respect to position were also observed for modelling the VCSPO in 1D and 2D geometry with results shown in 7 and summarised in 4. Again, the 2D model estimated a higher heat flux reduction magnitude with a maximum 62 % net boil off rate reduction possible. However, the P-O contribution to heat leak reduction is higher in the 1D model than for the 2D model. This, together with the different optimal VCS location indicates the impact of the VCS temperature assumption on the heat leak.

The addition of the P-O conversion slightly shifts the optimal 2D VCS position from $\chi_{VCS} = 0.47$ to 0.43 and from $\chi_{VCS} = 0.43$ to 0.33 for the 1D model. The shifting of the optimal VCS position towards the inner wall when including P-O conversion was also found by the 1D modelling studies of Xu et al., Meng et al. and Lv et al. [7, 9, 10]. These studies observed an optimal VCSPO position of $\chi_{VCS} = 0.4 - 0.5$ indicating that the optimal VCS position is a broad peak and generally consistent across different geometries and insulation types. The impact of the modelling assumptions and P-O conver-

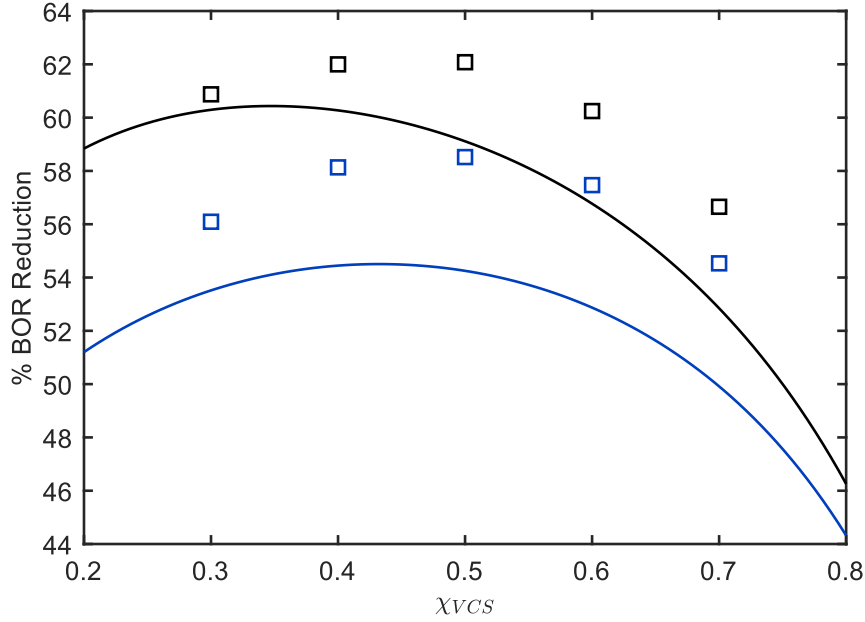


Figure 7: Steady state heat leak reduction of VCS and VCSPO with respect to radial position within insulation (χ_{VCS}).

sion extent on this result is subsequently discussed to assess the validity of the models used.

Table 4: Comparison of steady state LH₂ boil-off rates from a 40000 m³ tank with VCS in 1D and 2D geometry.

Net LH ₂ boil off (%/day)	1D model	2D model
No VCS	0.332	0.331
VCS	0.151	0.138
VCSPO	0.131	0.126
Optimal VCS position (χ_{VCS})	0.35	0.45
P-O contribution to VCSPO heat leak reduction	13.2	8.7

Both the 1D and 2D models describe a decreasing P-O contribution to the net heat transfer rate as the VCS position moved outward. This was due to the larger degree of sensible H₂ cooling as the VCS temperature increases at further out radial positions and the simultaneous decrease in the P-O conversion enthalpy above $T = 100$ K. This behaviour aligns with modelling

results from Shi et al. [5] where moving the VCS closer to the inner wall increased the contribution of the P-O conversion to the total VCS cooling duty. These results indicate that a catalytic VCS would have diminished P-O cooling utility in a LH₂ tank where the insulation has an ambient initial temperature such as during tank filling.

4.2. Impact of VCS model assumptions

The validity of assuming an axisymmetric temperature profile and complete P-O conversion was investigated to support selection of a VCS model for use in filling and transient modelling processes. Of these, the uniform temperature profile is the most significant assumption as it directly impacts the heat transfer driving force, the PUF thermal conductivity and the maximum p-H₂ composition change. Comparatively, the the assumption of complete P-O conversion would have less impact on VCS performance given P-O conversion only increases the heat leak reduction by appropriately 10 %.

The temperature profiles of the 1D and 2D VCSPO models are shown in 8. The 2D VCSPO temperature increases linearly from the inlet to the P-O converter at the VCS midpoint, after which it increases gradually. This phenomenon is not captured in the 1D model where the VCS temperature is approximately 8 K higher than the outlet of the P-O converter in the 2D model. The higher temperature in the 1D model results in more P-O conversion and the higher outlet temperature of the 2D model results in a smaller P-O cooling contribution compared to sensible heat as shown in 7. Together these factors result in the 1D model overestimating the P-O cooling contribution.

The assumption of complete P-O conversion did not significantly impact the P-O cooling contribution. In the 2D model, the equilibrium p-H₂ fraction was practically achieved at the outlet of the converter, however, there was a smaller p-H₂ fraction change compared to the 1D model due to the slightly lower outlet temperatures as shown in 5. The catalyst activity of the ferric oxide utilised achieved a nearly isothermal converter in 8 and indicates that for the purposes of VCSPO, development of enhanced activity catalysts would not offer significant benefits. For these reasons the assumption of incomplete P-O conversion is an insignificant assumption compared to the assumption of an antisymmetric temperature in the 1D model.

While the assumption of the axisymmetric temperature profile within the 1D model results in similar overall heat leak reduction estimates to the 2D model, it leads to appreciably different optimal VCS positions. Although

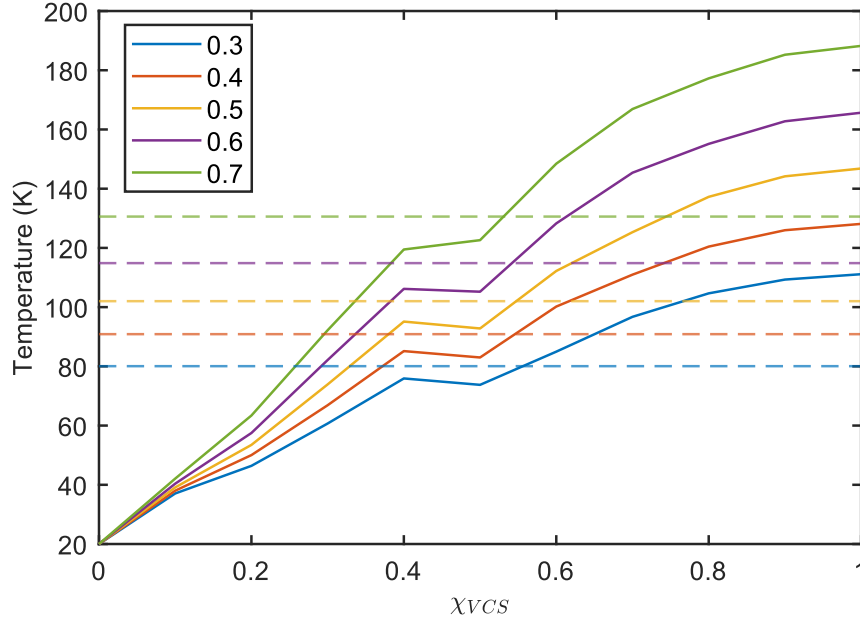


Figure 8: Temperature profile along the length of the VCSPO for different values of χ_{VCS} , where $\omega_{PO} = 0.5$. Solid lines represents the 2D model, dashed lines represent the 1D model.

Table 5: p-H₂ fraction change across 1D and 2D VCSPO models with respect to radial VCSPO position. Theoretical and actual p-H₂ compared for the 2D model

VCSPO position, χ_{VCS}	1D model	2D model	
	$X_1 - X_{2,e}$	$X_1 - X_{2,e}$	$X_1 - X_2$
0.3	0.512	0.384	0.383
0.4	0.574	0.472	0.471
0.5	0.619	0.542	0.541
0.6	0.657	0.607	0.606
0.7	0.688	0.663	0.662

this may not be an issue for high level thermal designs of VCS LH₂ tanks, the simplifications of the 1D model may lead to issues in subsequent rigorous VCS design stages. For the purposes of this study on investigating how VCS position impacts boil off reduction across filling, transient boil off and steady state stages, the 2D model was required to assess the impact of VCS position with high fidelity. Further, the use of the 2D model is also anticipated to

be more appropriate in the thermal and mechanical design of more complex VCS systems, where a larger temperature gradient along the VCS flow path may be present leading to different heat leak reduction and thermal stresses.

4.3. Losses during Chillydown

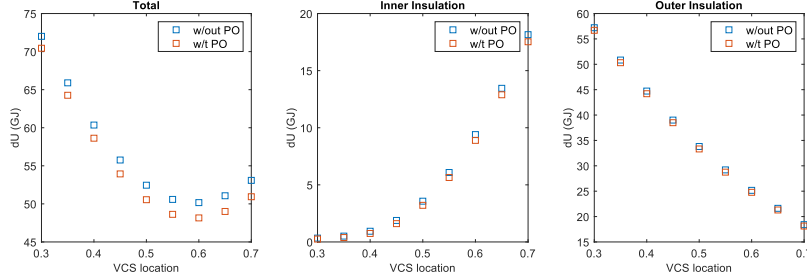


Figure 9

Figure 9 shows the residual thermal energy remaining in the insulation layer following cooldown of the tank inner wall, where the reference case is thermal steady state with the VCS. Unlike steady state BOR, where optimal χ_{VCS} was ~ 0.5 , the optimal VCS location for minimising residual energy was predicted to be ~ 0.6 for both VCS and VCSPO. While a small portion of this is attributed to the longer cooldown duration when χ_{VCS} was larger, this can be largely understood by considering the residual energy in the outer and inner insulation layers. When the VCS is placed close to the inner wall, the Since the specific heat capacity of PUF varies non-linearly (and is relatively low at lower temperatures), the residual energy of the insulation near the inner wall (where temperatures are lower) was generally

1. Not major difference between 50 and 70 % location
2. Very low specific heat capacity of the inner section is the main driver of it approaching zero, though we have to mention the reduction in mass
3. talk to the P/O reduction - relatively small overall contribution, interesting to note that the contribution is larger when vcs location is toward outer wall
4. look into what the dU is in reference to (VCS or no VCS)

Table 6 shows the cooldown duration with varying χ_{VCS} . Without the VCS, the cooldown duration was estimated to be 61.5 hrs. When the VCS was closer to the inner wall, there was generally a reduction in the time taken to cool the inner wall. This can be largely attributed to the more rapid cooling of the inner insulation layer. For $\chi_{VCS} = 0.3$, cooldown duration was 52.4 hrs. However, it should be noted that this benefit was only 7.8 % relative

Table 6: Comparison of steady state LH₂ boil-off rates from a 40,000 m³ tank with VCS in 1D and 2D geometry.

χ_{VCS}	0.3	0.35	0.4	0.45	0.5	0.55	0.6	0.65	0.7
VCS cooldown duration (hrs)	52.4	53.5	54.6	55.7	56.8	57.7	58.5	59.1	59.6
VCS-PO cooldown duration (hrs)	52.2	53.3	54.4	55.6	56.7	57.6	58.4	59.1	59.6

to the base case of $\chi_{VCS} = 0.5$. The addition of the para-ortho converter during the cooldown process was generally negligible, reducing cooldown duration by no more than 0.3%. This can be largely attributed to the fact that for the majority of the cooldown process, the temperature of the hydrogen vapour entering the VCS was too high for any significant para-ortho cooling to occur.

1. mention that the para ortho contribution is very small, due to the fact that most of the time, the vapour is coming out at 300 K

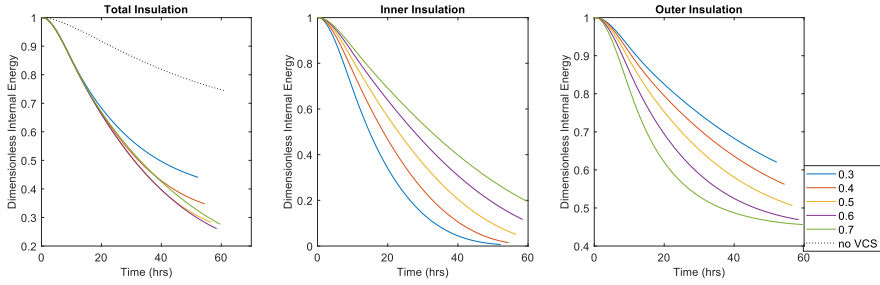


Figure 10

1. Non dimensionless?
2. Future work - looking at the effect of rate of filling.
3. Might need to include inner wall in addition to this?

4.4. Design Implications

Overall, the catalytic VCS indicated a net boil-off saving of 5740 kg/day for a 40000 m³ PUF insulated LH₂ tank. Assuming an optimistic cost of H₂ reliquification at \$2.5/kg (Al Ghafri, Munro et al. 2022) indicates a possible saving of \$14350/day through the reduced net boil-off rate. Incorporating the catalyst could reduce the net boil-off rate by 336 kg/day compared to the non-catalytic VCS and indicates a potential reliquefaction saving of \$840/day. While the P→O cooling contribution is modest, the lifetime vented LH₂ savings are likely to outweigh the costs of installing 649 L of catalyst in the VCS. The P→O cooling contribution could be marginally increased through employing multiple catalyst beds within the VCS flow path though increasing the activity of the Ionex catalyst is not expected to meaningfully increase the P→O cooling contribution for this application. VCS performance is likely to be strongly influenced by the costs of manufacturing, as a simple design with non-optimal performance may save more money than an optimal but costly VCS design though this trade-off is difficult to assess without VCS manufacturing cost estimates. Further work is required to estimate the cost of fabricating the VCS design within this study to verify the financial value proposition to support potential implementation of this technology

5. Conclusion

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Appendix A. Appendix